AMPX - Amprius Technologies | Deep Dive

Full active edition | Financial methods and portfolio Base overrides preserved.

Evidence update: 2 October 2026 | **Framework:** 8.0 | Historical company research: April-June 2026; saved financial anchor: Q1 2026 **Status:** NOT HELD in supplied workbook | **Classification:** inherited qualified high-energy-density battery thesis **Artifact analytical Base:** 46/60 =76.6667%, rounded **77** | **Protected Excel Base:** 75, unchanged **Active ceiling:** Q4 2027 **USD 34.650000/share**, using Q4 2027-Q3 2028 operating profit. This is the saved quarterly model, not the old annual FY2028 headline. **Valuation lock:** 25x forward-four-quarter operating profit,74M modeled diluted shares, unchanged factor 1 and USD unit factor 1. No default20x substitution.

Supercycle: DB7 Defense Battery + Physical AI Power + DB8 Defense

Core Thesis: The Physics of Flight

The product, customer and competitive descriptions in this inherited thesis reflect the June 2026 research date. They were not comprehensively reverified during this narrow funding-event update; the new agreement below does not verify every prior technical or market-leadership claim.

Ford Principle: not "batteries" - the physics of flight and autonomous mobility. Without 450+ Wh/kg, eVTOL doesn't fly profitably, drones don't complete missions, robots run out of power.

- **SiMaxx: 500 Wh/kg** - highest commercially available Li-ion on earth
- **SiCore: ~400 Wh/kg** - scalable, standard-manufacturing compatible
- **SA128: 320 Wh/kg** - shipping NOW, L3Harris validated
- Conventional graphite-anode Li-ion tops out at 250-300 Wh/kg. The 300→500 gap is physics, not engineering.

Dual moat (Axiom 14, the VICR parallel): physics (no competitor within 2x of 500 Wh/kg) + regulatory (NDAA Section 164 bans Chinese battery components from US defense). Chinese batteries are blocked by law AND physics simultaneously.

Customer Validation

L3Harris (first named defense prime) | Nanotech Energy (NDAA-compliant US contract manufacturing, SA128) | $21M Chinese electric-mobility order (SiCore) | eVTOL OEMs - Dubai commercial operations targeted year-end 2026 | HAPS operators.

Historical financial anchor and model limitations

The June 2026 research recorded the following historical values and then-current guidance: FY2024 $24.2M → FY2025 $73.0M (+202%) → FY2026 guided $125-135M (~75%). GM: 11% FY2025 → ~20% as of Q1 2026 - the trajectory is the evidence, and the level is the open question (moat test below). First EBITDA-positive quarter Q4 2025 ($1.8M). Cash $91.9M; manufacturing-light (>2 GWh via contract partners, minimal capex); runway then estimated at 18-24 months. These cash, guidance, margin and runway statements are dated historical anchors, not October 2026 verified current figures. The saved quarterly workbook identifies Q1 2026 actual revenueUSD28.5M and operating lossUSD6.7M. This pass does not perform a new earnings rescore or replace the share/cash assumptions.

Scoring breakdown: one specific catalyst point added

Category	Score	Evidence
Core Thesis	16/20	Physics-required above 350 Wh/kg; 500 Wh/kg unmatched; NDAA moat; L3Harris. Penalties: solid-state emerging (-2), margins thin (-1), concentration (-1).
Cycle Stage	7/10	Revenue tripled; first EBITDA; +75% guided. Stage 2 ramp; margins still thin.
TAM	9/10	Defense + eVTOL + robots + EV + soldier + space - universal energy-density primitive across phases.
Catalysts	5/5	Prior operating checkpoints retained; one point added for the specifically funded domestic production program announced in September. This is not a revenue or margin award.
Macro	9/10	NATO 5% GDP; drone warfare; NDAA mandate; Physical AI power demand.
Alpha	3/5	Silicon-anode physics under-appreciated; NDAA moat unmodeled; eVTOL TAM treated as speculative.
Risks	-3	GM ≤20% vs monopoly claim (-1). Solid-state pipeline (-1). Customer + Ukraine concentration (-1).
Artifact Base	**46/60 (77% rounded)**	Prior 45/60 plus one Catalyst point. Protected Excel Base remains75.

Project acCELLerate: restricted production funding

The September 23 agreement, filed September 28, supports domestic high-energy-density pouch-cell manufacturing for small UAS. It is a fixed-price **Other Transaction Authority prototype agreement**, explicitly distinct from a grant agreement or procurement contract. Its exact ceiling is **USD 74,793,173.11**, of which **USD 22.3M** is initially obligated and **USD 52,493,173.11** remains unfunded. Required company cost share is zero; performance and project costs are not zero.

Project acCELLerate: exact funding terms

Total contractual ceiling

USD74,793,173.11 | initial government obligationUSD22,300,000

Remaining appropriation

USD52,493,173.11 remains unfunded. Company cost shareUSD0.

Cash and ownership conditions

Accepted milestones precede payment. Equipment title is conditional and can vest in Amprius or its subcontractor after successful completion.

Source: September23 agreement, SEC Exhibit10.1, cover and sectionsVIII-IX; research transcription.

Research transcription of issuer evidence; not an issuer chart. Source: September23 agreement, SEC Exhibit10.1, cover and sectionsVIII-IX; research transcription.

Original agreement and issuer8-K. The base period runs September 23,2026 to September 22,2028. The payable milestone schedule, specific capacity, site and partner details are redacted. Amprius administers funds and subcontractor work; receipts cannot automatically be treated as retained profit or freely available cash.

Ledger field	Assumption and treatment
Currently funded scope	USD 22.3M government obligation; not cash already received
Completion / collection probability	75%, subjective judgment for that funded scope; execution and accepted milestones still matter
Conditional diagnostic receipts	USD 22.3M if completed and collected; probability-weighted diagnosticUSD16.725M

Ledger field	Assumption and treatment
Conditional eligible project outlays	Assumed equal to restricted receipts for this bounded bridge; weightedUSD16.725M outlays offset weighted receipts
Active operating-profit increment	USD 0. This is an explicit matching-cost modeling assumption, not proof the actual project margin will be zero
Additional appropriation	50% subjective probability on the USD 52.49317311M unfunded remainder; conditional sensitivity only, weighted funding amountUSD26.246586555M
Active cash balance	No new booked cash-ledger entry; milestone/outlay calendar cannot be isolated from existing manufacturing plans
Existing revenue overlap	Domestic production enables the existing defense battery vector. No separately disclosed battery purchase order or new sales pool is added
No double benefit	Funding is not both unrestricted income and the same amount of saved capex. Neither event probability discounts general battery revenue
Score decision	One Catalyst point:4 to5; artifact75 to77 rounded; protected Excel 75 unchanged

The matching-cost proxy recognizes that manufacturing funding pays for project work. It does not assert costless production, free profit, immediate liquidity or an actual zero project margin. Additional funding can reduce financing pressure only to the extent it finances otherwise necessary company outlays; capacity utilization, customer orders, retention and timing are unresolved. Conditional equipment title may vest in the subcontractor rather than Amprius, and funded equipment remains tied to the approved site. Either party may terminate for convenience with30 days notice and negotiated settlement. Government technical-data rights and domestic-participation requirements also constrain the program.

Probability diagnostic, not active cash forecast	0%	Active judgment	100%
Initial USD 22.3M funded-scope receipts	0.000M	16.725M at 75%	22.300M
Matching project-outlay assumption	0.000M	16.725M	22.300M
Incremental operating-profit bridge	0.000M	0.000M	0.000M
Additional USD 52.49317311M appropriation	0.000M	26.246587M at 50%	52.493173M

Raise or lower completion confidence on accepted milestones, commissioning, partner delivery and termination evidence. Revise the funding sensitivity when appropriations change. A future operating/cash change needs a bounded difference between new eligible receipts, existing planned spending, incremental project costs and resulting commercial output; absence of official forward profits is not itself an exclusion gate.

Business vectors retained in the saved total forecast

Defense UAS and drones remain the primary vector; eVTOL/UAM depends on certification and commercial deployment; electric mobility has its own customer/order exposure; humanoid robots remain an early opportunity; HAPS, space and soldier systems add possible high-density uses; legacy customers supply the starting base. The former annual product table was rounded thesis sizing, not a separately linked product model. It is replaced here with exact saved quarterly and annual totals, without inventing a new product allocation. The award is not a seventh additive sales vector.

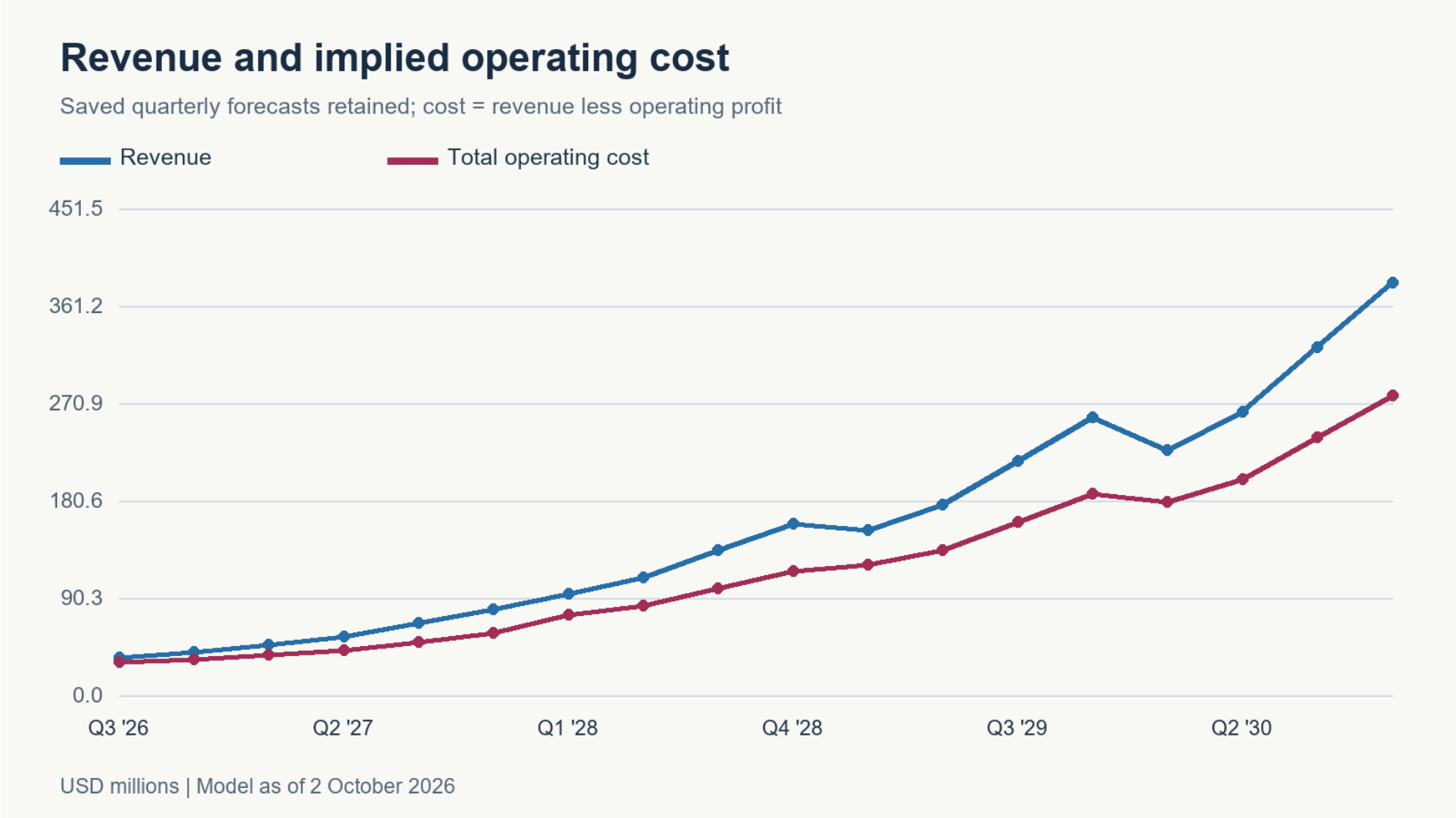
Current valuation and annual reconciliation

The active workbook preserves25x operating profit and 74M diluted shares. It derives each ceiling from the selected quarter and the following three quarters, retaining existing quarterly margins and seasonal allocations. No funding event changes those inputs. The old FY2028 USD 42 annual headline is superseded by the dated quarterly path below, not relabeled as a Q4 2027 target. The previously discussed25%annual operating-profit ambition remains a model assumption far above the historical gross-margin anchor, not a reported October achievement.

Calendar year	Saved revenue USD M	Saved OP USD M	OP margin	Shares M	OP multiple
2027	248.640	62.160	25.000%	74.000	25x
2028	497.280	124.320	25.000%	74.000	25x
2029	805.120	201.280	25.000%	74.000	25x
2030	1,195.840	298.960	25.000%	74.000	25x

Q4 2027 exact active chain: USD 102.564M forward-four-quarter OP x25 /74M shares = **USD 34.650000/share**. Forecast and control are identical because the evidence update adds zero operating profit.

Current saved quarterly model visuals

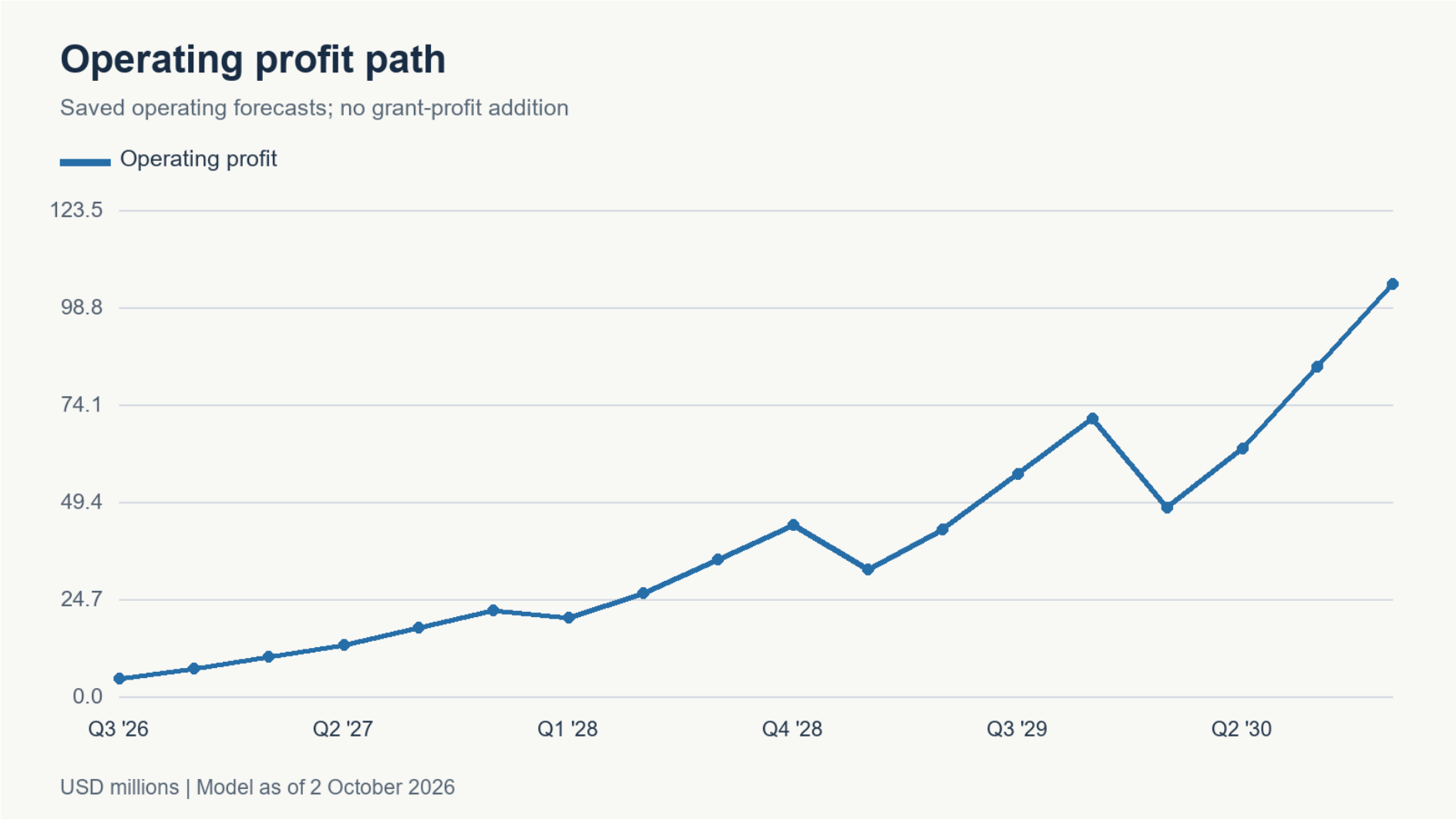


Revenue and implied operating cost. USD millions. Saved quarterly forecasts retained; cost = revenue less operating profit

Quarter	Revenue	Implied total operating cost
Q3 2026	35.431	30.956
Q4 2026	40.155	33.193

Quarter	Revenue	Implied total operating cost
Q1 2027	47.242	37.296
Q2 2027	54.701	41.647
Q3 2027	67.133	49.728
Q4 2027	79.565	57.809
Q1 2028	94.483	74.592
Q2 2028	109.402	83.294
Q3 2028	134.266	99.456
Q4 2028	159.130	115.618
Q1 2029	152.973	120.768
Q2 2029	177.126	134.858
Q3 2029	217.382	161.024
Q4 2029	257.638	187.190
Q1 2030	227.210	179.376
Q2 2030	263.085	200.303
Q3 2030	322.877	239.168
Q4 2030	382.669	278.033

Operating-profit bridge

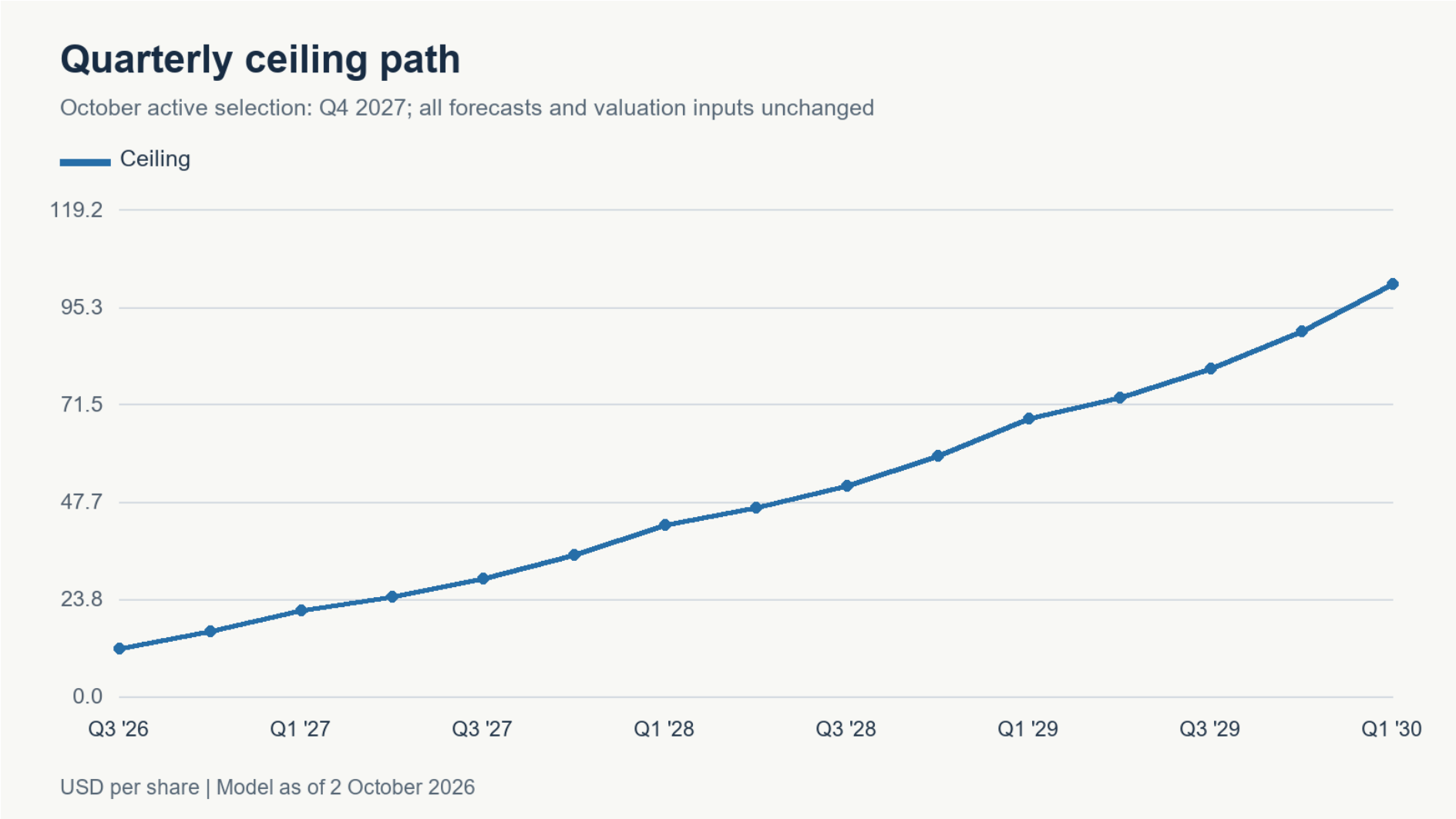


Operating profit path. USD millions. Saved operating forecasts; no grant-profit addition

Quarter	Operating profit	OP margin	Shares M
Q3 2026	4.476	12.632%	74.000
Q4 2026	6.962	17.337%	74.000

Quarter	Operating profit	OP margin	Shares M
Q1 2027	9.946	21.053%	74.000
Q2 2027	13.054	23.864%	74.000
Q3 2027	17.405	25.926%	74.000
Q4 2027	21.756	27.344%	74.000
Q1 2028	19.891	21.053%	74.000
Q2 2028	26.107	23.864%	74.000
Q3 2028	34.810	25.926%	74.000
Q4 2028	43.512	27.344%	74.000
Q1 2029	32.205	21.053%	74.000
Q2 2029	42.269	23.864%	74.000
Q3 2029	56.358	25.926%	74.000
Q4 2029	70.448	27.344%	74.000
Q1 2030	47.834	21.053%	74.000
Q2 2030	62.782	23.864%	74.000
Q3 2030	83.709	25.926%	74.000
Q4 2030	104.636	27.344%	74.000

Forward ceiling path



Quarterly ceiling path. USD per share. October active selection: Q4 2027; all forecasts and valuation inputs unchanged

Ceiling quarter	Forward four-quarter window	Ceiling / share
Q3 2026	Q3 2026 to Q2 2027	11.634000
Q4 2026	Q4 2026 to Q3 2027	16.002000

Ceiling quarter	Forward four-quarter window	Ceiling / share
Q1 2027	Q1 2027 to Q4 2027	21.000000
Q2 2027	Q2 2027 to Q1 2028	24.360000
Q3 2027	Q3 2027 to Q2 2028	28.770000
Q4 2027	Q4 2027 to Q3 2028	34.650000
Q1 2028	Q1 2028 to Q4 2028	42.000000
Q2 2028	Q2 2028 to Q1 2029	46.160000
Q3 2028	Q3 2028 to Q2 2029	51.620000
Q4 2028	Q4 2028 to Q3 2029	58.900000
Q1 2029	Q1 2029 to Q4 2029	68.000000
Q2 2029	Q2 2029 to Q1 2030	73.280000
Q3 2029	Q3 2029 to Q2 2030	80.210000
Q4 2029	Q4 2029 to Q3 2030	89.450000
Q1 2030	Q1 2030 to Q4 2030	101.000000

Key Risks

1. **GM vs monopoly claim** - at ~20%, the weakest margin profile of any chokepoint-classified name; revenue growth without margin expansion = competitive market, not monopoly
2. Solid-state pipeline (QuantumScape, Solid Power, Toyota) - AMPX holds a 2-4 year lead, not a permanent one
3. Customer concentration + **Ukraine ~35% of revenue** - single-conflict geopolitical exposure
4. No new US drone customers disclosed post-Q1 - possible domestic saturation
5. Cash runway 18-24 months - manufacturing-light helps; a raise is plausible before profitability

Re-Rate Triggers

1. **Historical GM moat test, registered May 2026 when artifact Base was75:** GM at or below20% in both Q2 and Q3 2026 was assigned a75-to70 analytical-score counter-trigger. Those results were not assessed in this narrow funding update; retain this checkpoint for a full earnings review. It never automatically changes the protected Excel Base.
2. GM printing ≥25% → margin-risk relief (+1 direction), monopoly claim validated with data

3. Dubai eVTOL commercial operations live (year-end 2026) → V2 MEDIUM→HIGH
4. New named US defense/drone customer → saturation concern resolved
5. Solid-state competitor commercial milestone above 400 Wh/kg → lead-erosion counter-trigger
6. Ukraine revenue share falling below 25% via diversification → concentration relief

Historical research retained from June 11,2026; recoverable pre-update original saved in the local run before folder. Funding evidence updated October 2; operating forecasts, valuation method and protected portfolio Base unchanged.